

DDTA BA3-T2 cross-baseline identity, staleness and lifecycle pressure test - R1

Status: COMPLETED / PROVISIONAL PASS WITH IDENTITY-LIFECYCLE REFINEMENT

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Executed microstep: BA3-T2 - cross-baseline identity, staleness and lifecycle pressure test

Scope discipline: this test does not close BA3, does not redesign BA2, does not define BA4 projections, AnalysisRecord/Common Finding, STRIDE overlays or ThreatForge implementation state.

1. Question under test

BA3-T1 proved that every BA identity needs baseline-scoped origin/provenance. T2 asks what happens when the governed baseline changes:

Which BA meanings remain the same identities, which must be reviewed, which are replaced, and which disappear?

The test deliberately uses concrete governed mutations instead of inventing a generic lifecycle state machine.

2. Candidate attacked

T1 left these questions open:

- cross-baseline identity/equivalence;
- accepted/rejected/stale/superseded/retired semantics;
- diagnostic resolution after source correction.

The pressure hypothesis was intentionally minimal:

H1 same source ID -> same BA identity

H2 one lifecycle rule can serve BReferent and BAProposition

H3 source revise can be copied as BA revise

H4 originState can carry acceptance/freshness meaning

All four hypotheses were attacked by the mutation corpus.

3. Test A - M1 Ethernet to Wi-Fi

3.1 Governed mutation

The facial-access corpus changes only D-3.5:

Ethernet available -> Wi-Fi available

Transport remains externally consumed; camera and recognition processor remain separated; FR-3.4 and its specialized constraints survive after review.

3.2 Identity pressure

The abstract LocalConnectivity meaning survives. The concrete realization does not.

If same document identity implied same BA proposition identity, the assertion:

```
realize(LocalConnectivity, WiredEthernet)
```

would have to mutate in place into:

```
realize(LocalConnectivity, WiFi)
```

That would make proposition identity denote two different normalized assertions over history.

3.3 Result

LocalConnectivity referent	RETAIN
WiredEthernet referent	RETIRE if unused elsewhere
WiFi referent	INTRODUCE
old realize proposition	REPLACE / SUPERSEDED
transfer/correlation/security propositions	RETAIN after review
same D-3.5 source identity -> same BA proposition	REJECTED

Finding A: source-document continuity and BA assertion continuity are independent decisions.

4. Test B - M2 external to project-owned transport

4.1 Governed mutation

D-3.4 changes from consumed external transport to project ownership/management.

4.2 Polarity pressure

BA2 represents the V0 boundary as a negative responsibility assertion. Project ownership changes that truth value.

Keeping one proposition identity while flipping polarity would violate assertion identity:

```
assignResponsibility[negative](project, transport, ownership/management)
```

becomes:

```
assignResponsibility[affirmative](project, transport, ownership/management)
```

4.3 Result

abstract transport/connectivity referent	RETAIN if meaning survives
negative responsibility proposition	REPLACE / SUPERSEDED
affirmative responsibility proposition	INTRODUCE as successor
external consumeService proposition	RETIRE if consumption ceases
transfer + specialized constraints	RETAIN after review
new project-owned transport branch	INTRODUCE as needed

Finding B: changing responsibility relations does not automatically split the referent whose responsibility changed.

5. Test C - M3 remote to local recognition

5.1 Governed mutation

Recognition moves onto the camera. The remote RecognitionCapture transfer disappears.

5.2 Retire versus replace pressure

A bad lifecycle rule would transform the old transfer proposition into any new local processing behavior merely because both concern recognition.

That would destroy the proposition's identity semantics.

5.3 Result

CameraSubsystem referent	RETAIN
RecognitionCapture referent	
RETAIN if acquisition meaning survives	
RecognitionProcessor	REVIEW / possibly RETIRE
remote delivery behavior referent	RETIRE
remote transfer proposition	RETIRE
old delivery-targeted constraints	
RETIRE old form unless successor authored	
D-3.3 placement proposition	
REPLACE when successor meaning is materialized	
D-3.4/D-3.5 related elements	
STALE FOR REVIEW where relevance was transfer-dependent	

Finding C: no successor is required merely because a neighboring function remains. **RETIRE** and **REPLACE** are materially different.

6. Test D - order-fulfillment internal versus external WMS

The order corpus states that **ADR-2.1** internal inventory authority is a genuine responsibility-boundary choice: an external WMS would replace many project-owned inventory FRs with integration/API/contract obligations.

This provides an independent domain control.

Expected BA effect:

- stable order/evaluation and governed result meanings may retain identity;
- internal inventory authority/capability/store meanings may retire or be replaced;
- internal mutation propositions retire/supersede;
- external service/contract referents and consumption/normalization propositions are introduced;
- payment and fulfillment meanings outside the affected boundary do not become stale merely because **MR-2** changed.

Finding D: mutation-local lifecycle handling generalizes beyond the camera corpus without requiring a domain taxonomy.

7. Family-specific identity tests

7.1 BAReferent test

Retain identity when the independently reusable project-semantic thing remains the same despite changed propositions around it.

Mutation examples forcing retention:

- **LocalConnectivity** across Ethernet/Wi-Fi;
- **RecognitionCapture** across remote/local recognition when capture itself remains required;
- order-evaluation/result semantics across provider-boundary changes where governed meaning is preserved.

7.2 BAProposition test

Retain identity only under normalized assertion equivalence.

A proposition identity therefore cannot survive a material change in:

- operator;
- polarity;
- role-bound participant identity/value;
- governed constraint value;
- local condition/temporal scope.

Source wording and provenance may change without forcing replacement.

Result: one shared retention rule for both BA1 families is rejected.

8. Review-state pressure

Origin state alone cannot encode review state.

Examples:

```
DERIVED + ACCEPTED
DERIVED + REJECTED
DIAGNOSTIC_UNRESOLVED + ACCEPTED
GROUNDED + STALE relative to new baseline
```

Therefore `PENDING_REVIEW | ACCEPTED | REJECTED` is required as a separate analytical review dimension.

This does not give BA project authority. It records whether analytical materialization is admitted to the accepted Base Analysis.

9. Staleness pressure

A new governed baseline does not invalidate all old BA meaning.

The minimum staleness rule is dependency-local:

```
changed source locator
-> directly grounded dependent BA elements need review
```

```
stale/replaced/retired derivation basis
-> derived dependents need review
```

```
replaced/retired proposition participant
-> proposition needs review
```

M3 also shows that effective context can matter even where the direct source file of an element did not change.

Therefore exact change-impact closure remains open, but **STALE** is required as a target-baseline revalidation state.

10. Supersession versus retirement

The mutation corpus forces both:

SUPERSEDED

A successor exists and takes over the earlier semantic responsibility.

Examples:

- Ethernet realization proposition -> Wi-Fi realization proposition;

- negative ownership proposition -> affirmative ownership proposition;
- remote placement proposition -> local placement proposition.

RETIRED

The old meaning ceases and no successor identity is required.

Examples:

- remote capture-transfer proposition after local recognition;
- delivery-behavior referent if no longer used;
- external service-consumption proposition after project ownership, if no external consumption remains.

Collapsing both into **inactive** would lose change explanation and targeted downstream invalidation.

11. Diagnostic resolution test

Suppose baseline B0 contains an accepted **DIAGNOSTIC_UNRESOLVED** element because sources conflict or are insufficient. B1 corrects the governed documentation.

Rejected behavior:

```
same diagnostic identity
originState: DIAGNOSTIC_UNRESOLVED -> GROUNDED
```

That rewrites history.

Accepted provisional behavior:

```
issue disappears      -> diagnostic RETIRE
new unresolved issue  -> diagnostic REPLACE / SUPERSEDE
new factual meaning   -> separate GROUNDED/DERIVED BA element
                        with own provenance
```

Thus origin remains historical while lifecycle expresses resolution.

12. Why BA-level REVISE is not required

The source corpus itself uses review labels such as **revise**, **revise/replace** and **retire/supersede** for governed documents.

Those labels are not a BA identity ontology.

At BA semantic identity level:

- non-identity metadata/provenance refresh with same meaning -> **RETAIN**;
- changed meaning with successor -> **REPLACE**;
- disappeared meaning -> **RETIRE**.

Adding **REVISE** would blur the exact boundary T2 is trying to protect: whether semantic identity survived.

13. Minimum candidate after pressure

Origin/provenance dimension:

```
GROUNDED | DERIVED | DIAGNOSTIC_UNRESOLVED
```

Review dimension:

```
PENDING_REVIEW | ACCEPTED | REJECTED
```

Freshness dimension:

CURRENT | STALE

Cross-baseline continuity disposition:

RETAIN | REPLACE | RETIRE

Derived historical interpretation:

accepted REPLACE -> SUPERSEDED

accepted RETIRE -> RETIRED

No new BAE family is forced.

14. Negative controls

- Do not make Git commit ancestry the semantic identity rule.
- Do not use document IDs as BA proposition identity.
- Do not preserve proposition identity solely because source wording is similar.
- Do not replace a referent merely because ownership/provider/realization changed.
- Do not auto-transfer retired security constraints to a new behavior.
- Do not mark the whole BA stale because one source file changed.
- Do not treat REJECTED analytical material as governed project truth.
- Do not mutate a resolved diagnostic into a grounded fact.

15. BA1 / BA2 reopen check

No mutation requires a third identity family or a new operator/role/modifier semantic.

The pressure is absorbed by:

- BA1 identities;
- BA2 proposition structure;
- BA3 provenance + continuity/lifecycle metadata.

BA1 reopen trigger: false

BA2 reopen trigger: false

16. Exit disposition

BA3-T2	COMPLETED / PROVISIONAL PASS WITH IDENTITY-LIFECYCLE REFINEMENT
BA3	STARTED / NOT CLOSED
BA1	REMAINS CLOSED
BA2	REMAINS CLOSED

The remaining BA3 work is narrower: derivation-rule reproducibility, effective dependency/change-impact lineage and feedback provenance, followed by closure review.

17. Next falsification target

Only after this package is committed, pushed and remotely verified:

BA3-T3 - derivation-rule reproducibility and change-impact lineage pressure test.

T3 should use the facial M4 effective-governed-context pressure plus provider-state normalization/order boundary cases. It must not start BA4 or method-specific analysis schemas.